



Upconversion luminescence of $\text{NaYF}_4: \text{Yb}^{3+}, \text{Er}^{3+}$ for temperature sensing

Shaoshuai Zhou, Kaimo Deng, Xiantao Wei, Guicheng Jiang, Changkui Duan, Yonghu Chen, Min Yin*

University of Science and Technology of China, Department of Physics, No. 96, JinZhai Road, Hefei, Anhui 230026, China

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ABSTRACT

Rare earth doped luminescent materials are promising candidates for temperature sensing. In this paper, pure $\beta\text{-NaYF}_4: 20\%\text{Yb}^{3+}, 2\%\text{Er}^{3+}$ microprisms were synthesized by hydrothermal method. The temperature dependence of the fluorescence intensity ratio (FIR) of emission bands centered at 525 nm and 545 nm was measured in the temperature range of 160–300 K under excitation of a 980 nm diode laser. The monotonous increase of FIR with temperature shows that this material can be used for temperature sensing. The dependence of the FIR on temperature is fitted with an exponential function and the effective energy difference obtained is 752 cm^{-1} , which gives further a temperature sensitivity of $1.20\% \text{ K}^{-1}$ at 300 K. The FIR technique was then used to study the effect of heating caused by laser excitation under various powers.

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1. Introduction

Temperature is one of the most fundamental physical parameters in many fields of science, engineering and industry. Conventional temperature sensors are based on the principle of liquid and metal expansion, in which the measurement of temperature is achieved by heat flow to an invasive probe [1]. However, due to the limitation of spatial resolution, sensitivity and accuracy of detection, such measurements are not suitable in many situations, such as the measurement of the fast-moving objects and intracellular temperature. Because of these restrictions, many studies have been focused on temperature sensing based on non-contact thermometry techniques, such as IR thermography, Raman spectroscopy [2], and luminescence [3]. Recently, non-invasive thermometers operating at sub-micron scales with high spatial and temperature resolution are a very active field [1,4–8]. The fluorescence life-time and the fluorescence intensity ratio (FIR) schemes, both of which were regarded as very promising techniques for temperature sensing [9], have received great attention [6,10–16]. In particular, FIR technique based on the temperature dependence of the FIR between two thermally coupled energy levels can improve the sensitivity and reduce the dependence of the measurement conditions, such as fluorescence loss, fluctuations of exciting light and electromagnetic compatibility problem [12].

With exceptional upconversion (UC) properties, Yb^{3+} and Er^{3+} co-doped fluorides have received great attention in many fields of application, such as bioimaging [17–19], nanothermometer [6], and

displays [20,21]. Among the investigated fluorides, $\text{NaYF}_4: \text{Yb}^{3+}, \text{Er}^{3+}$ was regarded as one of the most efficient UC materials and an ideal choice for FIR technique because of the suitable energy difference between $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ multiplets of Er^{3+} .

Compared with glass matrix for temperature sensing [16], nano and microscale powders are more likely to be applied in many situations, such as intracellular temperature fluctuations and temperature in microcircuits. Temperature dependent upconversion luminescence of $\text{NaYF}_4: \text{Yb}^{3+}/\text{Er}^{3+}$ nanocrystals was studied in Ref. [6], but a substantially different temperature dependence was reported more recently in Ref. [22]. This discrepancy is probably due to uncertain factors in the synthesis of nanoscale materials. So it is worth studying microscale powders carefully to obtain an accurate temperature dependence of FIR and study the relevant laser heating effect. In this work, the UC luminescence of micrometer-scale $\beta\text{-NaYF}_4: 20\%\text{Yb}^{3+}, 2\%\text{Er}^{3+}$ phosphor was studied at various temperatures under 29 mW excitation power intensity of a 980 nm diode laser for the purpose of temperature sensing. The specific co-doped concentrations ($20\% \text{Yb}^{3+}, 2\% \text{Er}^{3+}$) were selected due to the most intense UC luminescence [17,23]. The dependence of the integrated FIR on temperature was obtained and the experiment data can be fitted well with an exponential function. Moreover, the FIR technique was used to study the effect of heating caused by laser excitation under various powers, which can introduce obvious error if unsuitable excitation power was used.

2. Experimental

All of the chemicals are of analytical grade reagents and used without further purification. Rare earth nitrates standard solutions

* Corresponding author. Tel.: +86 551 3606912; fax: +86 551 3600817.
E-mail address: yinmin@ustc.edu.cn (M. Yin).

were prepared by dissolving the corresponding metal oxide (99.99%) in dilute nitric acid at elevated temperature.

β - $\text{NaYF}_4\text{:}20\%\text{Yb}^{3+}$, $2\%\text{Er}^{3+}$ was prepared by hydrothermal method [24] using ethylenediaminetetraacetic acid disodium salt (EDTA-2Na) as a chelating agent. The presence of EDTA-2Na, which is regarded as a good chelating agent, is found to be helpful to reduce the size and prevent further aggregation of the particles [24]. The $\text{RE}(\text{NO}_3)_3$ standard solutions ($\text{Y}/\text{Yb}/\text{Er}=78:20:2$, 2 mmol in total) were added dropwise to 8 mL aqueous solution containing 2 mmol EDTA-2Na resulting in a chelated RE-EDTA complex. After stirring for 30 min, 28 mL of aqueous solution containing 25 mmol of NaF was added into the above solution and stirred for 1 h. Then the complex precursor solution was transferred into a 50 mL Teflon bottle held in a stainless steel autoclave for hydrothermal treatment at 180°C for 24 h. Naturally cooled to room temperature, the precipitates were separated by centrifugation (4500 rpm, 5 min), washed with distilled water and ethanol for three times, respectively, and then dried in air at 60°C for 12 h.

Powder X-ray diffraction (XRD) pattern of the sample was performed on an X-ray diffractometer (Rigaku-TTR-III) with CuK_α radiation ($\lambda=0.15418\text{ nm}$) in the 2θ range from 10° to 70° . The size and morphology of the sample were obtained by a Field Emission Scanning Electron Microscope (FESEM, Model JSM-6700 F, JEOL, Japan). The upconversion emission spectra of $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} excited by a 980 nm diode laser were obtained by a Jobin-Yvon HRD-1 double monochromator equipped with a Hamamatsu R928 photomultiplier at various temperatures. The signal was analyzed by an EG&G 7265 DSP Lock-in Amplifier and stored into computer memories. The spot size of laser is about $1 \times 4\text{ mm}^2$. And the power of the exciting light was adjusted through neutral density filters. The powder sample was pressed into a tablet (thickness 0.8 mm, diameter 8.0 mm), and then was glued to the copper scaffold using cryogenic glue. Temperature of the sample was controlled over the range of 160–300 K by a WC50 helium compressor using copper as heat transfer media and a Lake Shore Model 321 temperature controller.

3. Results and discussion

3.1. Structure and upconversion luminescence

Fig. 1(a) shows XRD pattern of the as-prepared $\text{NaYF}_4\text{:}20\%\text{Yb}^{3+}$, $2\%\text{Er}^{3+}$ sample as well as the NaYF_4 standard data JCPDS No. 16-0334. Compared with the standard data of β - NaYF_4 , the sample exhibits the peaks of pure crystalline hexagonal phase. The general morphology and the structure of the sample are shown in Fig. 1(b). Regular hexagonal microprisms of a length roughly equal to $6\text{ }\mu\text{m}$ as well as diameter of $\sim 2\text{ }\mu\text{m}$ can be easily found from the SEM image.

Yb^{3+} ion is the most efficient and widely used sensitizer for 980 nm light for the UC phosphors based on Er^{3+} or Tm^{3+} , because of the strong absorption at 980 nm (whose absorption cross-section is 10 times larger than that of $\text{Er}^{3+}/\text{Tm}^{3+}$) [18]. To date, β - $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} is regarded as one of the most efficient UC phosphors for green emission [23].

The relationship of pump power and UC luminescence intensity at 539.6 nm was shown in Fig. 2. The UC luminescence intensity is in proportion to the n th power of the NIR excitation intensity, i.e., $I_{\text{upc}} \propto (I_{\text{NIR}})^n$, where I_{upc} is the UC luminescence intensity, I_{NIR} is the NIR excitation intensity, and n is the number of NIR photons absorbed in the UC process for one visible photon [25]. The slope of the fitting line is 1.95 at relatively low excitation density, indicating a two-photon UC process. The slope of the curve decreases at higher excitation density. This indicates the gradual saturation of UC process with the increase of excitation density.

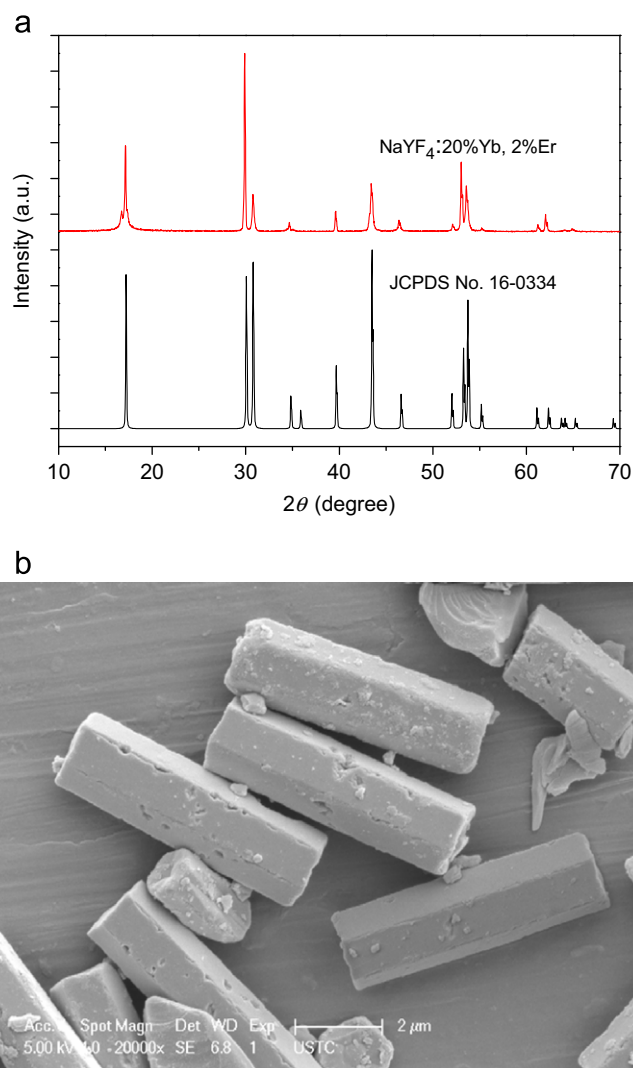


Fig. 1. (a) XRD pattern for the $\text{NaYF}_4\text{:}20\%\text{Yb}^{3+}$, $2\%\text{Er}^{3+}$ sample and the standard data of hexagonal β - NaYF_4 (JCPDS No. 16-0334) as a reference. (b) SEM image for the as-prepared $\text{NaYF}_4\text{:}20\%\text{Yb}^{3+}$, $2\%\text{Er}^{3+}$ sample.

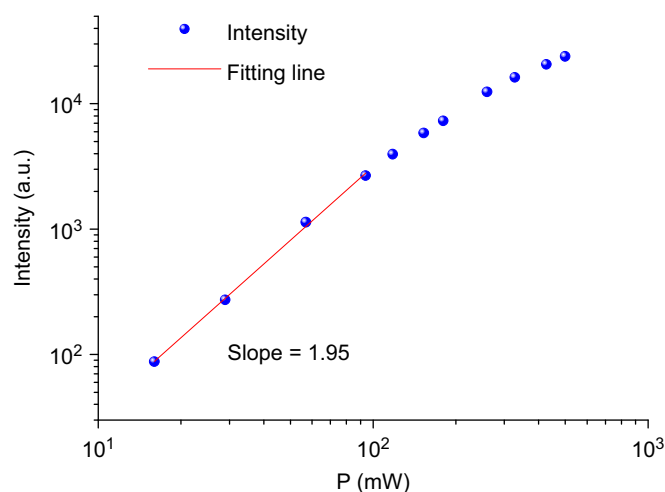


Fig. 2. Pump power dependence of $\text{NaYF}_4\text{:}20\%\text{Yb}^{3+}$, $2\%\text{Er}^{3+}$ upconversion intensity of 539.6 nm green emission peak. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Under 980 nm excitation, Er^{3+} ions are excited into $^4\text{F}_{7/2}$ multiplet through two energy transfer processes from neighboring excited Yb^{3+} ions. Then the $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ multiplets of Er^{3+}

are populated through a fast multi-phonon relaxation process followed by the transitions from these two multiplets to the ground multiplet $^4I_{15/2}$, respectively, emitting the dominant green UC luminescence [25]. In addition, the transition of $^2H_{9/2} \rightarrow ^4I_{13/2}$ can also be found in green emission region at relatively high exciting power, which is induced upon NIR excitation after $^2H_{9/2}$ population via a three-photon process [26].

3.2. Thermal effect

The $^2H_{11/2}$ and $^4S_{3/2}$ multiplets of Er^{3+} ion are closely spaced with a separation of only a few hundred wave numbers. At a certain temperature, a thermal equilibrium governed by the Boltzmann factor can be achieved quickly [6]. So the relative population of the two thermally coupled multiplets follows a Boltzmann distribution. Since the fluorescence intensity is proportional to the population of each energy level, the FIR of $^2H_{11/2}$ and $^4S_{3/2}$ emissions is given by [3,27]

$$R = \frac{I_{525}}{I_{545}} = B \exp\left(-\frac{\Delta E}{kT}\right), \quad (1)$$

where

$$B = \frac{\sigma_H \omega_H g_H}{\sigma_S \omega_S g_S} \quad (2)$$

g , σ , ω , I are degeneracy, the emission cross section, the angular frequency and the integrated intensities of the $^2H_{11/2} \rightarrow ^4I_{15/2}$ and $^4S_{3/2} \rightarrow ^4I_{15/2}$ transitions, respectively. ΔE is the effective energy difference between $^2H_{11/2}$ and $^4S_{3/2}$ multiplets; k is the Boltzmann constant; and T is the absolute temperature. The subscripts H and S stand for corresponding multiplets $^2H_{11/2}$ and $^4S_{3/2}$, respectively.

The relative change of the ratio R with respect to temperature variation S is a very important parameter for sensing application. According to Eq. (1), S can be written as,

$$S = \left| \frac{1}{R} \frac{dR}{dT} \right| = \frac{\Delta E}{kT^2} \quad (3)$$

This equation shows that, at a given temperature T , a larger energy gap between two thermal coupled energy levels gives a larger S , and a smaller temperature variation can be detected by measuring the change of intensity ratio.

Fig. 3(a) presents the green UC luminescence spectra of $NaYF_4:20\%Yb^{3+}, 2\%Er^{3+}$ under various temperatures from 160 K to 300 K using an unfocused laser beam of typically 29 mW at 980 nm. Low exciting power was utilized in order to reduce the heating effect caused by the excitation. As shown in Fig. 3(a), the green emission region consists of two distinct emission bands corresponding to the transitions from the $^2H_{11/2}$ and $^4S_{3/2}$ multiplets to the ground multiplet $^4I_{15/2}$, respectively. And the FIR of these two emission bands increased with the increase of temperature monotonously.

The fluorescence intensity of $^2H_{11/2} \rightarrow ^4I_{15/2}$ transition was integrated from 515.0 nm to 534.5 nm, while $^4S_{3/2} \rightarrow ^4I_{15/2}$ transition was integrated from 534.5 nm to 567.0 nm. The dependence of FIR on absolute temperature for $NaYF_4:20\%Yb^{3+}, 2\%Er^{3+}$ sample was shown in Fig. 3(b), which can be well-fitted with an exponential function:

$$R = 8.06 \exp(-1082.1/T) \quad (4)$$

And the effective energy difference obtained from the fitting is 752 cm^{-1} , which is close to the well-known splitting of $700\text{--}800 \text{ cm}^{-1}$ between $^2H_{11/2}$ and $^4S_{3/2}$ multiplets. Furthermore, the relative sensitivity S in the temperature range from 50 K to 500 K derived from Eq. (3) is shown in Fig. 4, and the S value is $1.20\% \text{ K}^{-1}$ at 300 K. It is noted that Eq. (4) is only used to show the

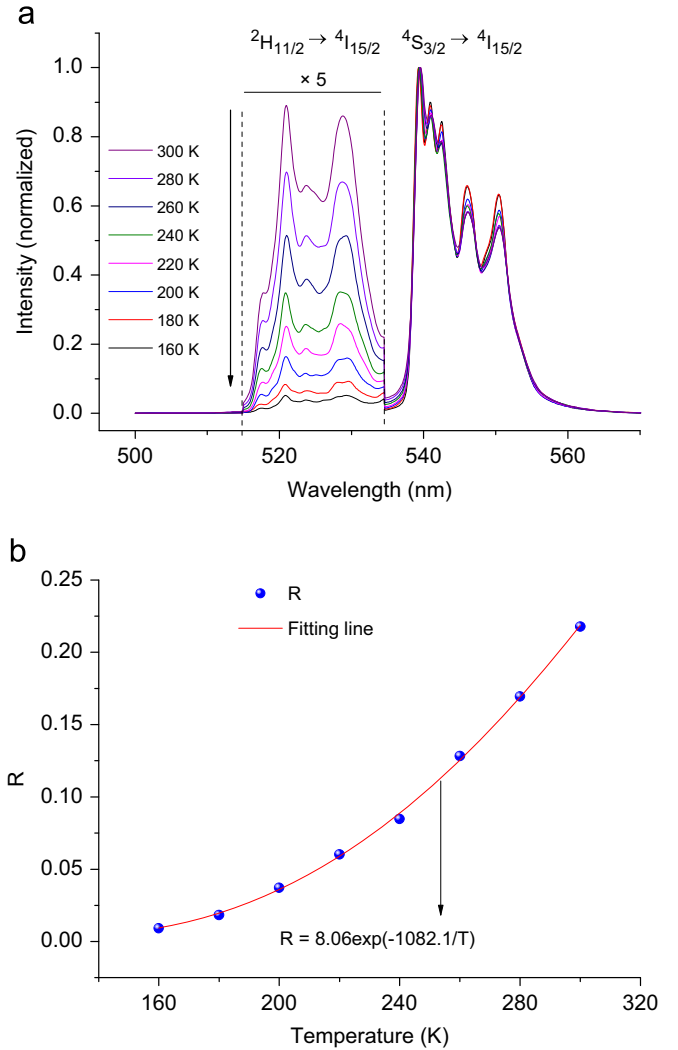


Fig. 3. (a) Temperature dependence of the green UC luminescence spectra of $NaYF_4: 20\%Yb^{3+}, 2\%Er^{3+}$ under 980 nm excitation. The spectra are normalized to the most intense emission peak at 539.6 nm. (b) FIR data versus absolute temperature for $NaYF_4: 20\%Yb^{3+}, 2\%Er^{3+}$ sample.

temperature dependence of R , and calibration is required in an actual application.

3.3. Heating effect by excitation

In order to investigate the heating effect caused by laser excitation. The UC emission spectra of the as-prepared sample were measured under various pump powers which were adjusted by putting different neutral density filters from 16 mW to 500 mW at three different sample chamber temperatures T_c (180 K, 240 K, and 300 K). As shown in Fig. 5(a), low excitation causes little influence on the UC spectra, however when the power is larger than 57 mW (14.25 mW/mm^2), the heating effect is not negligible. This phenomenon can be explained as follows. Owing to the strong absorption of 980 nm exciting light of Yb^{3+} ions and the transformation of part absorbed energy into heat energy by nonradiative processes, the concentration of Yb^{3+} ions (20%) has already caused obvious heating effect under a certain exciting power, leading to a deviation between the actual temperature of the sample and the sample chamber temperature. Recently, UC properties of $NaYF_4:Yb^{3+}/Er^{3+}$ phosphors with a wide range of Yb^{3+} concentration have been reported in Ref. [28], and they found that the two green emissions ratio increased with Yb^{3+} co-doping concentration monotonously.

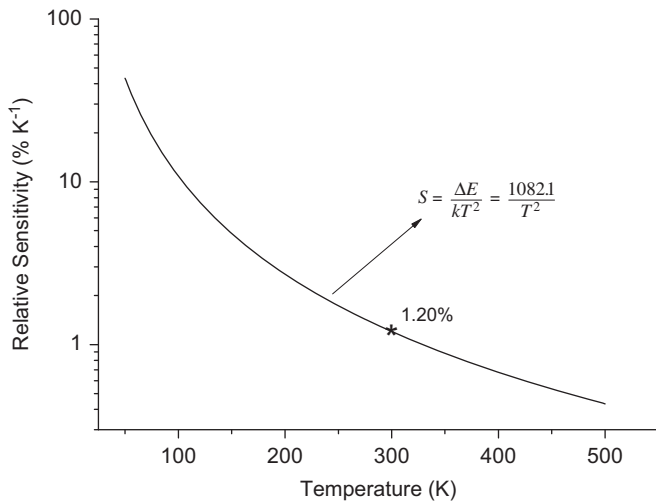


Fig. 4. Relative sensitivity in the temperature range of 50–500 K for NaYF₄: 20%Yb³⁺, 2%Er³⁺.

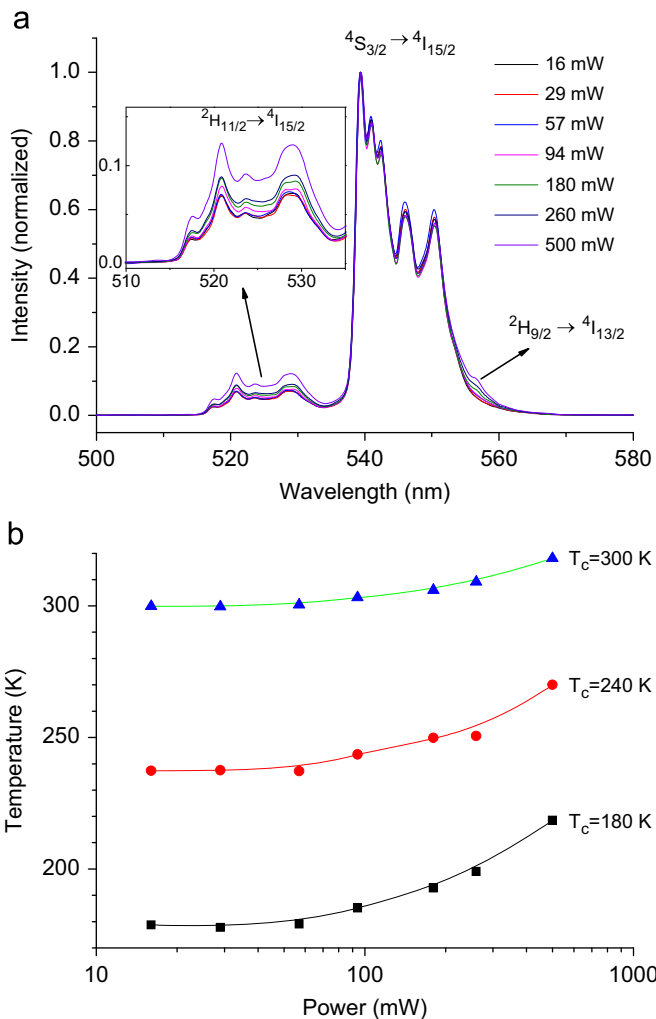


Fig. 5. (a) Normalized UC emission spectra of NaYF₄: 20%Yb³⁺, 2%Er³⁺ under different power of 980 nm excitation at 240 K. The inset is the amplified picture of the first emission band. (b) The relationship between the actual temperature of the sample and the exciting power at various sample chamber temperatures T_c . The spot size of laser is about 1×4 mm².

Considering both the fluorescence intensity and the heating effect, 29 mW was selected to excite the sample to obtain the dependence of FIR on temperature in this work.

Fig. 5(b) shows the deviation between the sample chamber temperature and the actual temperature of the sample which was obtained from the fitting Eq. (4) under different laser power intensities. It was found that the deviation at low temperature was larger than that at high temperature. When the sample chamber temperature is 180 K, 240 K, 300 K under 500 mW exciting power, the calculated deviation is 38 K, 30 K, and 18 K, respectively. In addition, a weak emission peak located at 557 nm originated from the $^2H_{9/2} \rightarrow ^4I_{13/2}$ transition can also introduce influence under high exciting power, which can be easily found in Fig. 5(a). And this influence will increase obviously with the increase of the exciting power intensity due to its three-photon UC process nature.

4. Conclusion

In summary, regular hexagonal β -NaYF₄:20%Yb³⁺, 2%Er³⁺ microprisms were synthesized by hydrothermal method. And UC luminescence spectra of the as-prepared sample were investigated at various temperatures from 160 K to 300 K under a 980 nm diode laser beam of 29 mW. The FIR of $^2H_{11/2} \rightarrow ^4I_{15/2}$ and $^4S_{3/2} \rightarrow ^4I_{15/2}$ transitions and temperature was well-fitted with an exponential function and the obtained effective energy difference between $^2H_{11/2}$ and $^4S_{3/2}$ is 752 cm⁻¹. Based on the obtained theoretical curve, the relative sensitivity of the FIR to temperature is obtained as 1.20% K⁻¹ at 300 K. The heating effect was studied by measuring the FIR under various excitation laser powers. The result shows that when the power intensity is larger than 14.25 mW/mm², the heating effect is not negligible for the sample studied. Under a suitable exciting power density, based on a relative deviation of 0.5% in FIR measurable by an ordinary spectrometer, a variation of 0.4 °C at ambient temperature can be resolved using FIR calibrated to a series of temperatures.

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